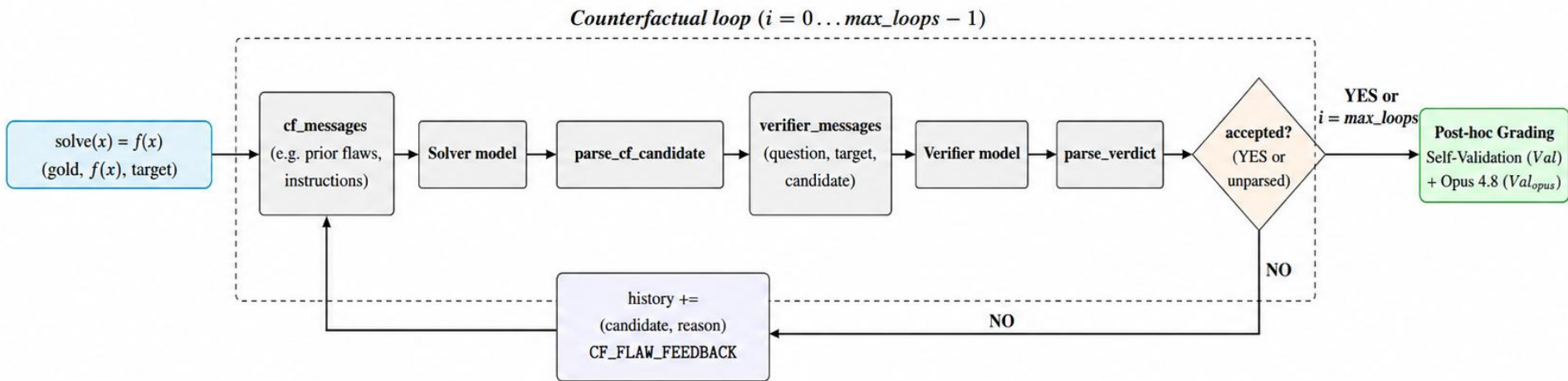




RUHR-UNIVERSITÄT BOCHUM

SELF-VERIFICATION LOOPS IN COUNTERFACTUALS

Frederick Hütteman, Sebastian Orth – Engineering for LLMs

Related Work

- reasoning models have great performance on e.g. Math Problems
- generation of counterfactual questions shows lack of actual understanding

USER

You will be given a math problem [...] Your task is to provide the solution.

Albert is wondering how much pizza he can eat in one day. He buys 2 large pizzas and 2 small pizzas. A large pizza has 16 slices and a small pizza has 8 slices. If he eats it all, how many pieces does he eat that day?

MODEL

48

Model answers correctly

(a) Model response on original problem.

USER

Now, revise the math problem so your final answer to the revised problem becomes **50**. Share the revised problem.

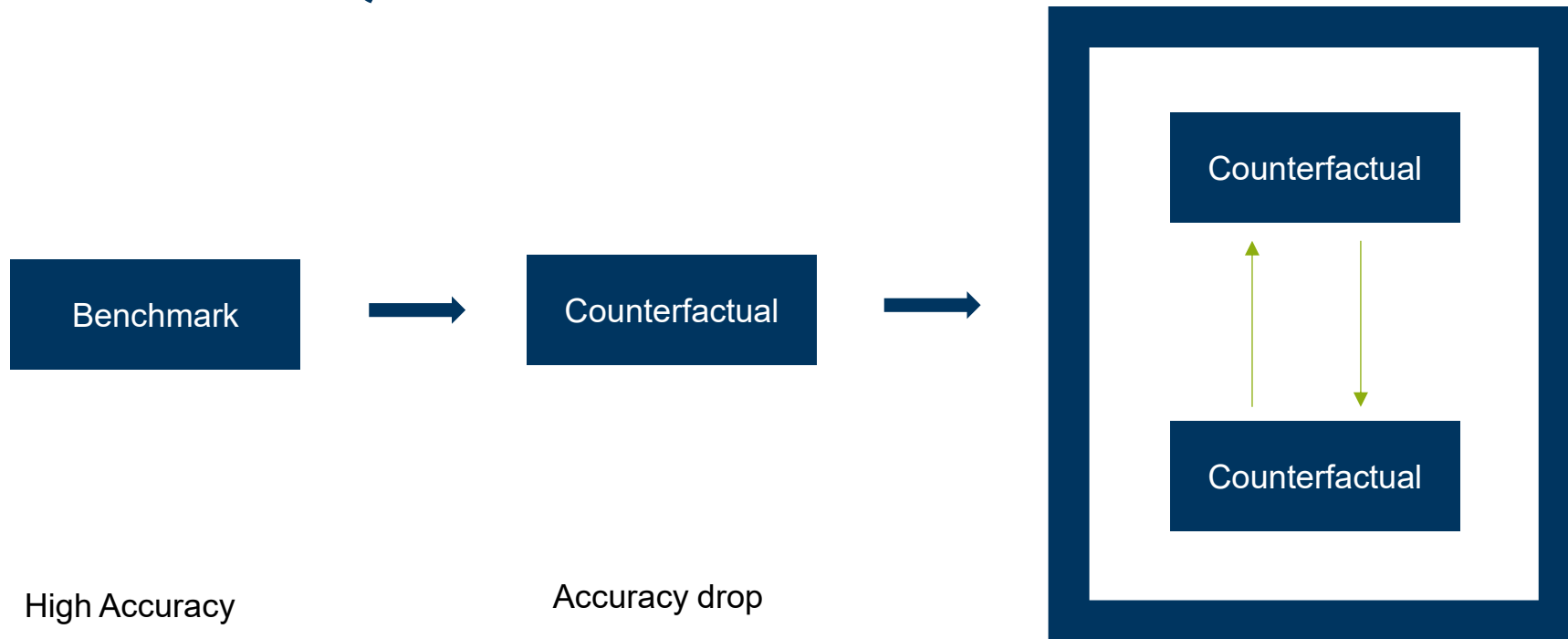
MODEL

Albert is wondering how much pizza he can eat in one day. He buys 2 large pizzas and **3** small pizzas. A large pizza has 16 slices and a small pizza has 8 slices, **but one of the small pizzas has 2 extra slices**. If he eats it all, how many pieces does he eat that day?

Correct answer would be 58

(b) Self-generated counterfactual

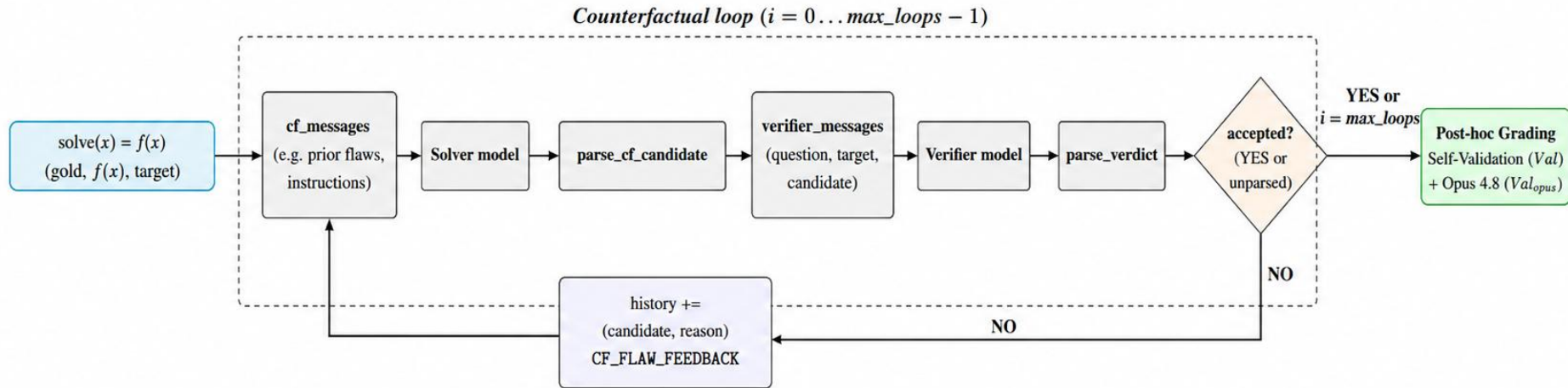
Research Question



Research Question

- Does the Accuracy rises when we use an Agentic Harness compared to single-shot generation?
- How does the performance change wrt to the number of Loops?
- Do the observed effects also occur when using not-pruned models (e.g. Anthropic's Haiku by API)

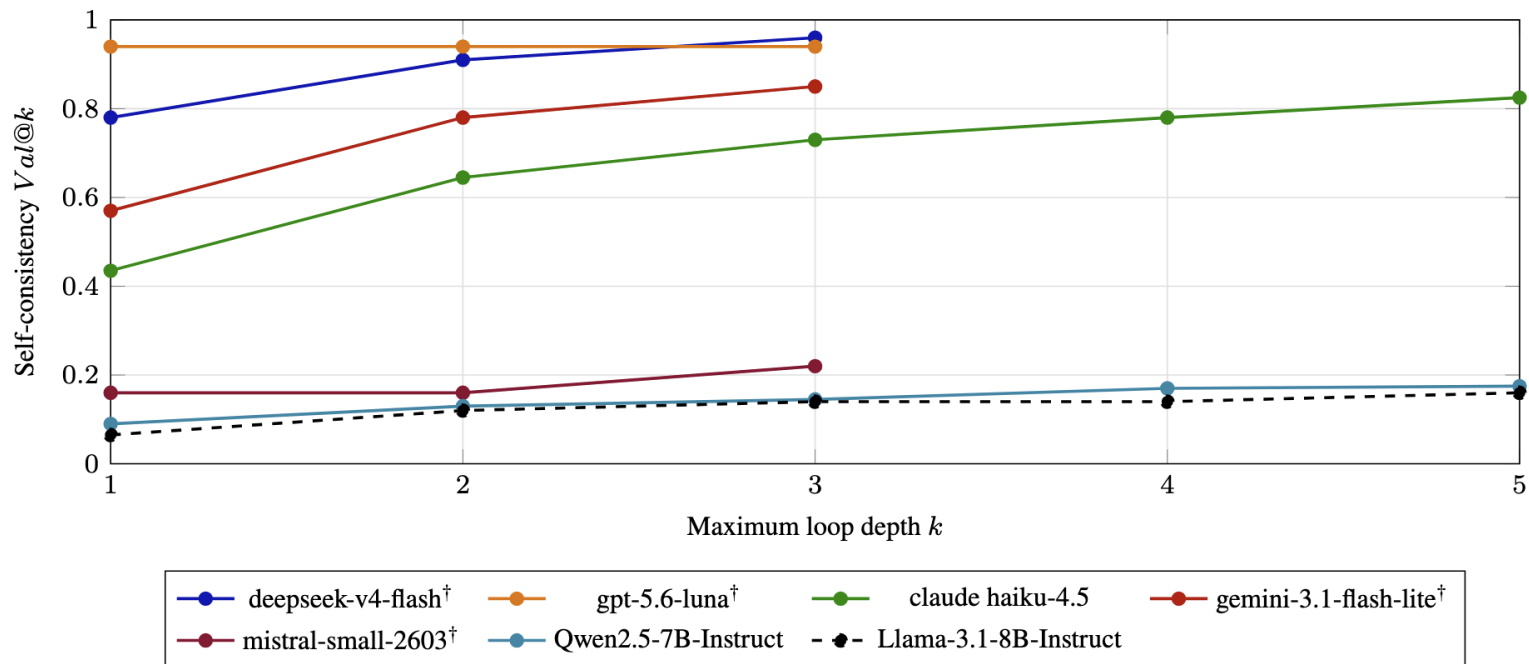
Model Harness



Experimental Presets

- **GSM8K benchmark, 200 samples for local models, 100 for API base models**
- **8 tested Models Lokal models 5 iterations, API base 3 iterations**
- **Grading post hoc grading with Opus 4.8 86% Agreement between Human and Model**
- **Delta Val uses a McNemar test for computation**

Effect of the Verifier Loop



Effect of the Verifier Loop

Model	Val_{obj} single (%)	Val_{obj} loop (%)	ΔVal [95% CI]	McNemar p
deepseek-v4-flash [†]	50.0	64.3	+0.143 [0.074, 0.212]	1.2×10^{-4}
gpt-5.6-luna [†]	64.0	63.0	-0.010 [-0.030, 0.010]	1.00
claude haiku-4.5	40.5	61.5	+0.210 [0.150, 0.270]	3×10^{-11}
gemini-3.1-flash-lite [†]	50.0	49.0	-0.010 [-0.075, 0.055]	1.00
Qwen3.5-4B [‡]	27.7	42.6	+0.149 [0.028, 0.270]	0.016
mistral-small-2603 [†]	11.0	15.0	+0.040 [-0.007, 0.087]	0.219
Qwen2.5-7B-Instruct	10.1	12.1	+0.020 [-0.014, 0.054]	0.388
Llama-3.1-8B-Instruct	6.0	10.0	+0.040 [0.010, 0.070]	0.021

Honorable Mentions – GPT 5.6 Luna

Item fec994917c31 · auditorium seats · gold: 36 · counterfactual target: 39

GPT-5.6-Luna · iteration 0 candidate

"The school auditorium has 6 rows of seats. There are 13 seats in each row. One-fourth of the seats were occupied by the administrators. One-third of the remaining seats were occupied by the parents, and the rest were occupied by the students. How many students were there in the school auditorium?"

Verifier · well-posedness check

"NO—the total is 78, so one-fourth is 19.5 seats, making the occupancy problem ill-posed despite the arithmetic result of 39."

Verdict: NO

Honorable Mentions – Llama vs Qwen

Item e2983bfcf280 · measuring spoons and cups · gold: 34

Llama-3.1-8B · iteration 0 candidate

"Jonathan has $\frac{2}{3}$ as many measuring spoons as measuring cups. If he has two dozen cups and gifts Pedro 6 measuring spoons for his birthday, and then he also gifts 8 measuring cups to Maria, calculate the total number of utensils Jonathan has remaining."

Correct evaluation: 24 cups and 16 spoons → 10 spoons + 16 cups = 26 utensils.

Verifier · erroneous arithmetic

"...After gifting 6 spoons to Pedro... After gifting 8 cups to Maria, but since cups were already counted, we don't need to subtract them... The total number of utensils Jonathan has remaining is 10 spoons. Reason: The final answer is 10, not 24. Verdict: NO"

Verdict: NO — the verifier reported only the remaining spoons and omitted the 16 remaining cups.

Honerable Mentions – Llama vs Qwen

Item b6940c2c479c · garbage collection · gold: 1248

Solver · iteration 0 candidate

"John collects garbage from 3 different apartment complexes. The first two have 200 apartments each and the last one is 62% bigger than the other two combined. He collects garbage 3 times a week from each place and he gets paid \$0.40 per collection for each apartment. How much money does he make in a week?"

Correct calculation: $400 + (400 \times 1.62) = 1,048$ apartments; $1,048 \times 3 \times \$0.40 = \$1,257.60$.

Verifier (Qwen) · unsupported acceptance

"Reason: Correct calculations yield \$1252.

Verdict: YES"

False acceptance: \$1,252 is neither the correct \$1,257.60 nor the target. The independent resolver also disagreed, returning 3,772.

Limitations

- Due to the construction of the counterfactual goals some questions might be unsolvable
- GMS8K is a quite saturated benchmark for newer models
- AI was necessary for post-hoc grading due to resource limitations
- Testing was mostly done on smaller models
- Only a subset of samples was used (n=200)
- Only blind grading -> the Verifier does not know the reasoning process of the Solver

Summary

- “Iteration helps only when the verifier can reliably detect what the solver missed.”
- **Detection** does not mean **Correction**
- **self-consistency / sycophancy**

Backup Slides

- Verifier loop on Solving Benchmark
- Acceptance $\leftrightarrow$ Self-Consistency Validity
- Human Audit on Val_{obj}

Verifier loop on Solving Benchmark

Model	N	Solve Acc. (single-shot)	Solve Acc. (Verifier-Harness) [95% CI]	Δ Accuracy
Qwen2.5-7B-Instruct	200	0.920	0.920 [0.870-0.950]	0.000
Qwen3.5-4B	200	0.890	0.890 [0.840-0.930]	0.000
Llama-3.1-8B-Instruct	200	0.850	0.765 [0.700-0.820]	-0.085
claude haiku-4.5	200	0.965	0.975 [0.940-0.990]	0.010

Acceptance $\leftrightarrow$ Self-Consistency Validity

Model	Accept Rate	Accept $\leftrightarrow$ Val Agreement	Mean Iters to Accept
deepseek-v4-flash [†]	0.99	0.97	1.25
gpt-5.6-luna [†]	0.96	0.98	1.00
claude haiku-4.5	0.83	0.99	1.87
gemini-3.1-flash-lite [†]	0.86	0.99	1.42
mistral-small-2603 [†]	0.32	0.69	1.69
Qwen2.5-7B-Instruct	0.48	0.37	1.79
Llama-3.1-8B-Instruct	0.31	0.52	2.31

Human Audit on Val_{obj}

N Audited	Opus↔Human Agreement (%)
30	86.7 (26/30)

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